

Enhancing dynamic ECG heartbeat classification with lightweight transformer model

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ABSTRACT

Arrhythmia is a common class of Cardiovascular disease which is the cause for over 31% of all death over the world, according to WHO's report. Automatic detection and classification of arrhythmia, as an effective tool of early warning, has recently been received more and more attention, especially in the applications of wearable devices for data capturing. However, different from traditional application scenarios, wearable electrocardiogram (ECG) devices have some drawbacks, such as being subject to multiple abnormal interferences, thus making accurate ventricular contraction (PVC) and supraventricular premature beat (SPB) detection to be more challenging. The traditional models for heartbeat classification suffer from the problem of large-scale parameters and the performance in dynamic ECG heartbeat classification is not satisfactory. In this paper, we propose a novel light model Lightweight Fusing Transformer to address these problems. We developed a more lightweight structure named LightConv Attention (LCA) to replace the self-attention of Fusing Transformer. LCA has reached remarkable performance level equal to or higher than self-attention with fewer parameters. In particular, we designed a stronger embedding structure (Convolutional Neural Network with attention mechanism) to enhance the weight of features of internal morphology of the heartbeat. Furthermore, we have implemented the proposed methods on real datasets and experimental results have demonstrated outstanding accuracy of detecting PVC and SPB.

1. Introduction

Cardiac arrhythmia is a significant class of cardiovascular disease (CVD) [1]. Arrhythmias can be classified into two categories. The first category includes arrhythmia that is formed by a single irregular heartbeat, which is known as pathological arrhythmia. The second category includes arrhythmia that is formed by a series of irregular heartbeats, which is referred to as rhythmic arrhythmia. Certain features of arrhythmias, such as the painlessness and insignificance of silent myocardial ischemia, may cause many sudden deaths, thereby requiring the long-term monitoring of specific patients [2]. Therefore, the continuous monitoring of patients with several CVDs, including post-myocardial infarction, tachycardia, arrhythmias, and left ventricular dysfunction, plays a key role in the effective assessment of

cardiovascular health [3].

An electrocardiogram (ECG) records the physiological state of the heart over a period. One of the primary tasks that implement the dynamic monitoring of ECG signals is optimal for collecting ECG in real time. At present, many methods are used to measure/record ECG: in-the-person, on-the-person, and o-the-person [4]. Most of the household devices that are used for ECG measurement belong to the on-the-person category. An ECG signal includes a number of heartbeats, each consisting of a group of successive waves. These waves can reflect the health information of different parts of the heart.

With the improvements in hardware information transmission and computing capabilities, wearable ECG devices are becoming an important diagnostic approach [5]. As wearable ECG monitoring devices are commonly based on single-lead measurements with dry metal plates

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[6,55], they can result in much smaller signal amplitudes and noisy waveforms compared to measurements with adhesive electrodes [7]. In general, the strength of a dynamic ECG signal from wearable devices is weak and easily interfered with by various noises, artifacts, and other human electrical signals. Moreover, the type and intensity of the signal noise vary substantially under different conditions, which presents challenges for signal analysis and the auxiliary diagnosis of diseases. Therefore, it is crucial to reduce and eliminate noise [8] to improve the signal quality of dynamic ECG [9]. In particular, studies have proved that the accuracy rate is less than 80% [10] when several more advanced complex QRS detection models are tested on the dynamic noise dataset. To deal with these issues, we use a Butterworth bandpass filter to remove noise from the baseline drift, and wavelet transform and a soft threshold to remove electromyographic interference.

In recent years, deep learning approaches have been used to detect arrhythmia. One of the rationales behind this detection method for arrhythmia is that various types of arrhythmia exist, each of which is associated with a pattern, so that it is possible to recognize and classify arrhythmia using traditional and deep learning approaches. Current deep learning methods include two basic convolutional network models: the convolutional neural network (CNN) and recurrent neural network (RNN). The CNN is one of the most popular deep learning model architectures, which uses gradient-based optimization algorithms for training [11,12]. For example, a one-dimensional (1D) CNN can be used to integrate the two main parts of ECG signal feature extraction and arrhythmia classification [13]. Furthermore, it automatically learns the appropriate ECG feature representation from the original ECG data, thereby eliminating the need for handmade features.

The RNN is another deep learning model that is most suitable for learning sequential input and time-series data. At each step of learning, the hidden units in the RNN receive input data, update the hidden state, and finally, make predictions [14,15]. For example, a global and updatable classification scheme known as the global RNN (GRNN) [16] can explore the potential features of an ECG signal based on the morphological features and time-dependent features of the ECG signal.

The Transformer is also a popular model that was originally used in natural language processing (NLP) [17,18,19]. Two of the excellent features of the Transformer are that it only uses the attention mechanism, without using complex convolutional operations, and it can encode the location information and integrate all dependencies into matrix operations to reflect the timing information. Because the ECG signal reflects the heartbeat and pumping of blood, which is also time sequential [20,21], a similar mechanism to the Transformer can be applied to ECG signals.

However, existing deep learning approaches such as the CNN, RNN, and Transformer still exhibit several limitations. First, most of the existing dynamic ECG classification models have large-scale parameters for training, which results in poor performance in real dynamic environments. Second, the attention mechanism in the Transformer that is used to model long-range dependencies faces great challenges owing to the context size, such as the quadratic complexity in the input length. Moreover, existing algorithms based on the Transformer have a large number of parameters, so that they cannot be directly applied to wearable ECG devices with low power consumption.

To address these issues, we propose a novel model Lightweight Fusing Transformer based on Transformer model for real wearable ECG devices. To apply Transformer to ECG time series, we have made the following changes. We implement the following changes to apply the Transformer to ECG time series: First, we use the encoder part only, because ECG signals do not have translation signals. Second, we propose a CNN-based input structure to enhance the dynamic ECG feature acquisition capability. Subsequently, to solve the large-scale parameter problem, we implement a more concise and powerful structure in the form of a two-level attention mechanism: a local attention mechanism in the word embedding, and the global attention mechanism in LightConv. The experimental results show that our method can reduce 72% of the

parameters and ensure good performance.

Our main contributions are as follows.

- We propose a novel model Lightweight Fusing Transformer based on Transformer model for the real wearable ECG devices. In particular, we develop a more concise and powerful structure in the form of a two-level attention mechanism: the local attention mechanism in the word embedding and the global attention mechanism in LightConv to solve the large-scale parameter problem. This model can effectively detect premature ventricular contractions (PVCs) and supraventricular premature beats (SPBs) with fewer parameters and high accuracy.
- We propose a novel CNN-based input structure to enhance the feature extraction capability from dynamic ECG signals further. Moreover, we use a Butterworth bandpass filter to remove baseline drift noise, and use wavelet transform and a soft threshold to remove electromyographic interference to overcome the low data quality problem. Our approaches enhance the performance of the proposed model on dynamic ECG.
- We conducted an extensive experimental evaluation using real datasets. Our experimental results demonstrate that the developed techniques outperform previous approaches in terms of the accuracy of the PVC and PVB detection. The results indicate that the proposed methods can yield acceptable results and can be deployed in wearable device environments.

The paper is organized as follow. Section 2 provides the related work and the background information. Section 3 describes the proposed framework, including data processing, segmentation and model details. In Section 4, we demonstrate our experiments and evaluation results.

2. Related work

Generally, the algorithm based on traditional methods of ECG heartbeat classification includes four steps: (1) pre-processing of original ECG signal data; (2) QRS complex detection and segmentation of ECG heartbeat; (3) feature extraction of ECG signals; (4) building classifier based on signal. Recently, the research trend has shifted to the deep learning method.

The first stage of heartbeat classification can also be called signal processing. Traditional signal processing technology has been developed for several decades. FIR/IIR high-pass filtering and wavelet transform, as common methods, have been used to reduce motion artifacts [22,23,24]. Specifically, the most commonly used method in current practice is the IIR high-pass filter with a cut-off frequency of 0.5/0.67 Hz [25,26]. This has laid the foundation for the subsequent work.

In the second stage, heartbeat segmentation is an indispensable process [27] because the object of general arrhythmia detection is a single heartbeat. As the QRS complex is the band with the strongest energy in the ECG signals, the usual heartbeat segmentation methods are carried out around it [28]. The most commonly used method is the adaptive detection threshold nonlinear translation algorithm (Pan-Tompkins algorithm) proposed by Pan and Tompkins [29], which is simple and easy to implement. In a recent paper [30], sensitivity of Discrete Wavelet Transformation (DWT) to detect QRS complex is 99.87%, and the detection error rate is 0.42. Exciting results have been achieved in the detection of R peaks using empirical mode decomposition (EMD) technology and digital filters [31].

In the field of traditional arrhythmia detection, in order to further improve the detection effect, especially for the class S, researchers began to use the attention mechanism in the field of arrhythmia detection and have achieved good research results. Kiranyaz et al. [32] proposed the use of an adaptive one-dimensional convolutional neural network to integrate feature extraction and classification to achieve a fast and accurate real-time heartbeat classification model based on specific patients. Experiments on MIT-BIH show that excellent classification

Table 1
Detail information of training data.

Recordings	Length (h)	N beats	V beats	S beats	Total beats
A01	25.89	109,062	0	24	109,086
A02	22.83	98,936	4554	0	103,490
A03	24.70	137,249	382	0	137,631
A04	24.51	77,812	19,024	3466	100,302
A05	23.57	94,614	1	25	94,640
A06	24.59	77,621	0	6	77,627
A07	23.11	73,325	15,150	3481	91,956
A08	25.46	115,518	2793	0	118,311
A09	25.84	88,229	2	1462	89,693
A10	23.64	72,821	169	9071	82,061

performance has been achieved in the detection of positional beats (Class V) and supraventricular ectopic beats (Class S). Zhang et al. [33] proposed a convolutional recurrent neural network (STA-CRNN) based on spatial and temporal attention. The network consists of a CNN sub-network, a spatial and temporal attention module, and an RNN sub-network. It was used to focus on Heartbeat features of spatial and temporal dimensions. In order to solve the problem that CNN cannot accept variable length ECG signals, Yao et al. [34] proposed an attention-based time incremental convolutional neural network (ATI-CNN), which integrated CNN, recursive networks and attention modules to achieve ECG for signal spatiotemporal information fusion. The current state-of-the-art model (as far as we know) is the Long Short-term Memory (LSTM) model that combines local attention and global attention which our team proposed in 2020 [35]. Natarajan [36] et al. combined static hand-crafted ECG features with deep features extracted from the raw ECG waveform via a neural network. Learned embeddings were fed into a Transformer architecture. The research of Yan et al. [37] only used the encoder of the transformer and combined the characteristics of the R-R interval to achieve better results, and the model has good parallel execution performance.

Arrhythmia detection of wearable devices is mainly an extension of methods based on models of ordinary environments. More considerations and innovations were used in noise reduction and signal compression, as well as model simplification [38,54]. Hu et al. [39] used the ECG analogy front end and digital processing on the node to be designed to eliminate most of the noise and deviation. A new hierarchical hidden Markov model was also proposed. Xia et al. [40] designed a stacked denoising autoencoder (SDAE), and used softmax regression to achieve real-time detection. Amirshahiet al. [41] employed spike-timing dependent plasticity (STDP), and reward-modulated STDP (R-STDP), in which the model weights were trained according to the timings of spike signals, and reward or punishment signals. Azariadi et al. [42] used Discrete Wavelet Transform (DWT) and Support Vector Machine (SVM) classifier to analyse the ECG signal. Saadatnejad et al. [43] proposed a new architecture composed of wavelet transform and multiple LSTM recurrent neural networks.

3. Proposed framework

In this section, we propose a novel model named Lightweight Fusing Transformer that can automatically identify abnormal heartbeat from healthy control subjects using dynamic ECG signal data. The proposed processing framework includes several components: data acquisition, removing noise and heartbeat segmentation and reshape, which will be discussed in the following sub-sections.

3.1. Data acquisition

We use the dynamic single-lead ECG recordings [44] as our dataset for heartbeat classification and arrhythmia detection. This dataset consists of 10 dynamic single-lead ECG recordings collected from arrhythmia patients by using a unified wearable ECG device with a

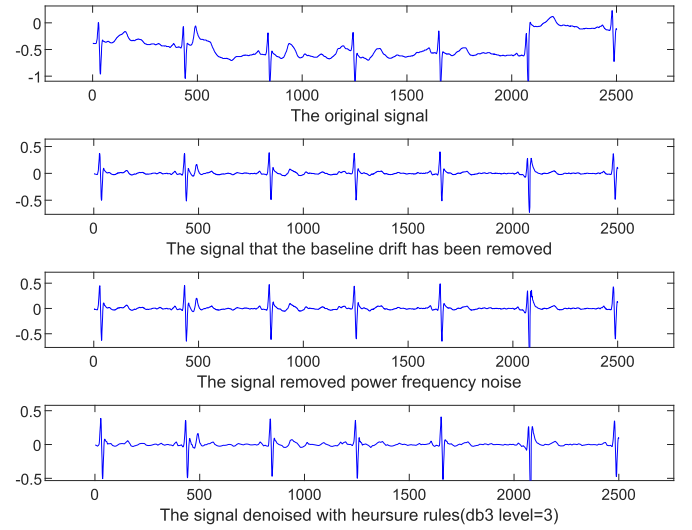


Fig. 1. Noise reduction process.

sampling frequency of 400 Hz. Each of the recording lasts for about 24 h (shown in Table 1). In addition, this dataset also includes premature ventricular contraction (PVC) and supraventricular premature beat (SPB) with low signal quality and abnormal rhythm waveforms. As this dataset comes from real wearables and is much larger than the well-known MIT-BIH dataset, it is more appropriate to our research.

3.2. Removal of baseline drift and electromechanical noise

As dynamic ECG signals that are collected from wearable devices often include noise that is generated from the environment, measurement or contact with the human body, we first need to remove noise from the raw data, which includes the removal of baseline drift and electromechanical noise.

3.2.1. Baseline drift

Baseline drift is a common type of ECG signal noise that is usually caused by low-frequency interference (at a frequency less than 5 Hz), such as electrode movement or breathing. As a result of the noise, the signal baseline deviates from the normal position in the diagram and the overall signal fluctuates up and down. In Fig. 1, signal A indicates the original signal with baseline drift and signal B is the processed signal. As the mixing of low-frequency noise and actual low-frequency information can seriously affect the signal quality, it can lead to incorrect waveform recognition, thereby strongly affecting the diagnosis and evaluation of the disease. To address this issue, we use a Butterworth bandpass filter to deal with baseline drift.

A signal processing filter with a at frequency response in the pass-band is known as a Butterworth filter (maximum at amplitude filter) [45]. The Butterworth filter calculation formula is as follows:

$$H(j\omega) = \frac{1}{\sqrt{1 + \varepsilon^2 \left(\frac{\omega}{\omega_p} \right)^{2n}}} \quad (1)$$

where n denotes order, $\omega = 2\pi f$, ε denotes the chase with gain (A_{max}). The cut-off frequency selected in this study is 5 Hz, which is also considered as the best cut-off frequency for ECG signal processing [46].

Fig. 1 presents the changes in several signals before and after noise reduction. It can be observed from the figure that the waveform tended to be stable and was centered around the central axis. However, after removing low-frequency noise, the Q wave became smaller and the ST had an additional waveform. Therefore, we used discrete wavelet transform to process the signal.

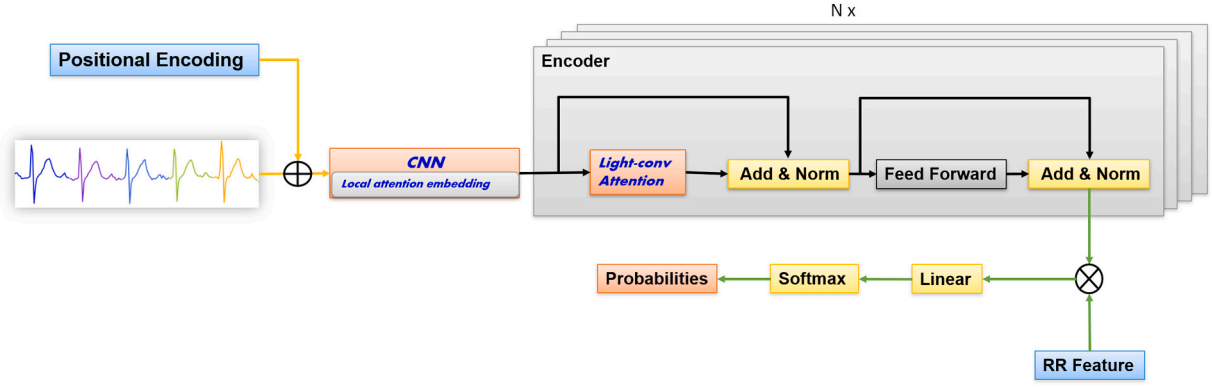


Fig. 2. Based on the original model [37], the model improves the input Embedding and self-attention. The new model uses CNN with a local attention mechanism as embedding. A novel convolution structure is designed after Embedding in place of self-attention.

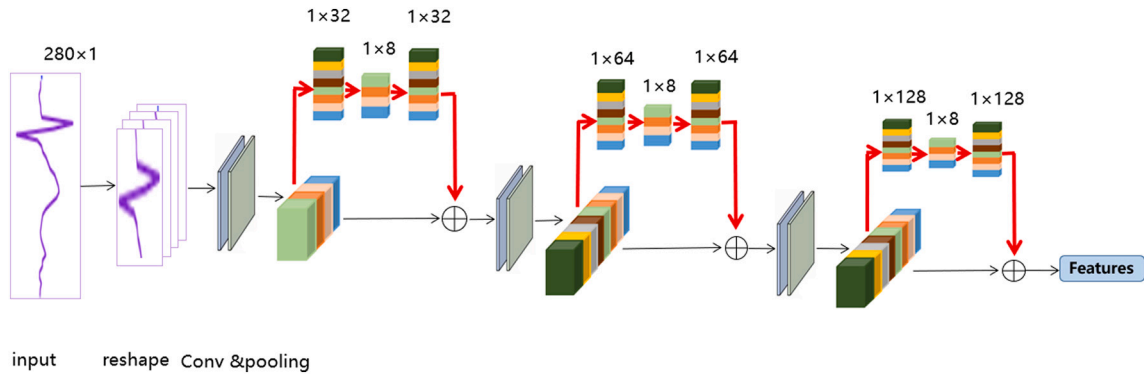


Fig. 3. Architecture of CNN Input embedding with local attention. The part connected by the red line is the flow chart of local attention mechanism. The eigenvectors are first sequenced to obtain the connection between the signal points and extended to add the weight of each channel of the eigenvectors. Then, each weight is multiplied by each channel in the original eigenvector. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

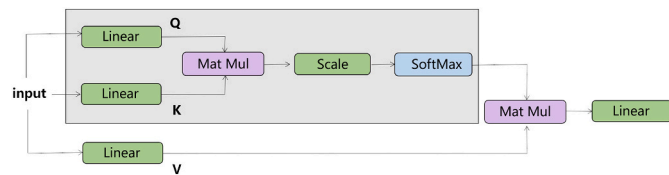


Fig. 4. The self-attention architecture of Transformer. Q (Query), K (Key) and V (Value) are derived from the same input. First, calculate the dot product between Q and K and divide it by a scale $\sqrt{d_k}$, where d_k is a dimension of Query and Key vectors. The Softmax operation is used to normalize the result into a probability distribution, and then multiplied by the matrix V to obtain the representation of the weight sum.

3.2.2. Electromechanical interference

Abnormal fluctuation of the ST segment occurred while smoothing the signal (Figs. 1–3). To eliminate electromechanical noise, we applied wavelet transform and a soft threshold to process and combine the signals. It can be observed from it can be seen from Figs. 1–4 that after reconstruction, the fluctuation of the ST segment disappeared, whereas the overall signal became more regular and smooth. In particular, we applied the heursure rules.

$$\text{eta} = \frac{\|x\|^2 - n}{n} \quad (2)$$

$$\text{crit} = \frac{[\log(n)/\log 2]^{1.5}}{\sqrt{n}} \quad (3)$$

$$\lambda = \begin{cases} \lambda_1, \text{eta} < \text{crit} \\ \min(\lambda_1, \lambda_2), \text{eta} \geq \text{crit} \end{cases} \quad (4)$$

where n denotes length of $x(t)$, λ_1 is the threshold by threshold rule and λ_2 is the threshold based on unbiased likelihood estimation rule.

3.3. Heartbeat segmentation and reshaping

We first used the appropriate algorithm to detect the R-peak position of each heartbeat, and then split ECG signal segments and finally standardized each beat.

3.3.1. R-peak detection

Only the labels of the S and V R-peak points were provided in the dataset, which was not sufficient for heartbeat segmentation and heartbeat classification. In general, the sensitive QRS complex detection algorithm [47,48,49] consists of three main stages: bandpass filtering, generating potential blocks, and thresholding. In this case, we first used Two Moving Average algorithm [47] to detect the position of R-peak in ECG signals.

1. Bandpass filtering: We used bandpass filtering in the QRS detection, and employed a second-order Butterworth filter with different selected pass bands for accurate QRS detection in the time domain.
2. Generating potential blocks: Based on the normal duration of the QRS interval which is 100 ± 20 ms for a healthy adult, we generated potential wave blocks. Furthermore, we used two moving averages for the onset and offset of the potential QRS waves in the ECG signals.

The first moving-window integration was used to capture the QRS area and it was the threshold for the output of the second moving-window integration.

$$MA_{QRS}[n] = \frac{1}{W_1} (y[n - (W_1)] + y[n - (W_1 - 2)] + \dots + y[n]). \quad (5)$$

where $W_1 = 44$ which is the window width of QRS segment. The second moving-window integration was used to capture a complete beat.

$$MA_{Beats}[n] = \frac{1}{W_2} (y[n - (W_2)] + y[n - (W_2 - 2)] + \dots + y[n]). \quad (6)$$

where $W_2 = 231$ is the window width of a complete heartbeat.

When the amplitude of the first moving average filter was greater than that of the second moving average filter, that part of the signal was selected as a block of interest.

3. Thresholding: Weave blocks that were smaller than the expected width of the QRS complex were rejected. The rejected blocks were noisy blocks and the accepted blocks contained R-waves. The maximum absolute value within each accepted block was considered to be the R-peak.

3.3.2. Segmentation

As we conducted an arrhythmia test for each heartbeat and raw signals are successive heartbeats, the heartbeat segmentation was an essential task. After obtaining the results of the R-peak detection, we intercepted the same length of time before and after the R-peak point and summarized the two lengths as the length of each segment. Owing to individual differences, the segment time length of each patient was set to be slightly longer than the time length of the heartbeat. For example, if the position of a certain R-peak was 2000, the segment was intercepted as [1700, 2200]. However, all subsequent segments were filled or downsampled to a fixed 280 points. The fragment length of 10 patients was as follows: 280, 290, 270, 280, 300, 300, 240, 230, 300, 260.

3.3.3. Normalization

Heartbeats with different sequence lengths should be reshaped and standardized. For this purpose, each beat was compressed or filled to 280 data points. We first used cubic spline interpolation to regularize and standardize the heartbeats. Cubic spline interpolation obtains the curve function set by solving the equations of three bending moments through a smooth curve with a series of shape and value points. More precisely, this approach standardizes the data based on the mean and standard deviation of the original data. For example, the original value x of A can be standardized to x' using the Z-score. The Z-score in particular is more applicable to the case where the maximum and minimum values of attribute A are unknown or there are outliers beyond the value range.

3.4. Model details

Our model is based on Fusing Transformer [37], which ameliorates the Transformer architecture. The Transformer is a well-known sequence-to-sequence model in the NLP field with encoder and decoder parts. However, this model can also be applied to the medical field. To use the Transformer in ECG signals, the Fusing Transformer eliminates the decoder to adapt to the specific application scenario of medical fields because ECG signals lack translation. To apply the Transformer to our application effectively, we make several changes to the input embedding and self-attention. First, we propose a new CNN-based input structure to enhance the feature extraction capability from dynamic ECG signals further, as opposed to the 1D CNN in the original model. Second, to solve the large-scale parameter problem, we propose a more precise and powerful structure to replace the self-attention part in the original model. Moreover, we embed the two-level attention mechanism into the model: the local attention

mechanism in the word embedding and the global attention mechanism in LightConv.

Given a sequence $X = (x_1, x_2, \dots, x_T)$, where T denotes the length of the given heartbeat, Transformer outputs a vector $P = (p_1, p_2, p_3)$, where p_i denotes the probability of the sequence judged to class i . Fig. 2 depicts the architecture of the proposed model.

3.4.1. Input embedding

Input embedding is the first step in our model. In the Transformer, word embedding is an important step at the beginning of the model. To map the points at each location to numbers, a structure is proposed to replace the word embedding in [37]. The structure uses a 1D convolutional layer, and the input and output of the layer are the same size. However, there is only one convolutional layer, which limits its ability to extract intra-heartbeat features. To address this issue, we propose a CNN-based input structure to enhance the capability of intra-heartbeat feature extraction from dynamic ECG signals. In our new model, the first level of attention (a local attention mechanism) is embedded into a CNN module that learns the local morphological features from each heartbeat in the input. Fig. 3 presents the proposed attention architecture.

First, we design our new model for each 1D individual heartbeat sequence, which is reshaped into a local channel feature space $X = (x_1, x_2, \dots, x_T) \in \mathbb{R}^{1 \times N \times C}$, where C denotes channels and the size of each channel is $1 \times N$. The space mapping of reshaping involves putting the heartbeat internal local band stacking on the channel dimension ($N = 10, C = 28$).

Second, we input X into three structures with channel attention mechanism (local attention mechanism) sequentially, and then output an eigenvector F with a local morphological feature relationship in the heartbeat. Specifically, there is a 1D convolution layer F_{conv} and a maximum pool F_{max} in the first two structures. Finally the convolution features after pooling are fed through a channel attention mechanism $F_{attention1}$.

The third structure in Fig. 3 contains the channel attention mechanism, but does not contain the maximum pooling layer. To improve the ability to extract the intra-heartbeat features from the beat, our model performs a series of effective operations. First, the convolution features after pooling are recalculated through the F-channel attention mechanism. Second, the local morphological features within the heartbeat are weighted. After the nonlinear correlation between the local morphological features is learned, the correlation between the local morphological semantics is enhanced. All of these operations enable the model to emphasize the features of individual wavebands within the beat (e.g., P, QRS, and T bands with distinct features) and to suppress several invalid features (e.g., noise in the dynamic ECG signal and certain baseline features).

Furthermore, to emphasize the effective local features and to suppress invalid features, we use the channel attention mechanism to weight the convolutional features after pooling. This mechanism can improve the sensitivity of the model to local morphological information in the heartbeat. As illustrated in Fig. 3, the output feature of the maximum pooling after each convolutional layer is followed by a branch, namely the channel attention mechanism (SENet) [50].

In this framework, an F_{GAP} global average pooling (GAP) and two consecutive Full Connection (FC) layers are included in each branch structure. After a series of operations in the branch structure, the weight, $S \in \mathbb{R}^{1 \times C}$ of each channel corresponding to feature U' , gives the weight S to the feature U' through the weighting operation. All of these new structures and operations realize the weighting of the local semantic features in the heartbeat, increase the attention to the useful local waveforms in the heartbeat, and enhance the correlation between the local features. Moreover, to further use of correlation between U' channel, convolution F_{conv} and pooling F_{max} essentially involves a fusion of two dimensions to obtain U' , rather than a spatial correlation. Finally, the global average pooling F_{GAP} is used to block the space features of

correlation, which features $1 \times \frac{N'}{2}$ in spatial dimensions and generate a feature at the channel level.

$$Z = F_{GAP}(U') = \text{Concat}_{i=1}^{C'} \left(\frac{1}{M} \sum_{j=1}^M u_j' \right) \in \mathbb{R}^{1 \times C'} \quad (7)$$

We reduce dimension for feature Z , then increase the dimension of the features, where F_{fc1} sequences the $C'Z$ of the channels to $\frac{C'}{r}$ (r is compression ratio) to reduce calculation.

$$S = \sigma(W_2 \delta(W_1 Z^T)) \in \mathbb{R}^{C' \times 1} \quad (8)$$

We then use a rectified linear unit (ReLU) to learn the nonlinear correlation between the local semantics within the heartbeat and use a sigmoid function to learn the relationship between the local morphology within the heartbeat, rather than the role of distinct local features in the classification of heartbeats.

$$\delta(z) = \max(0, z) \quad (9)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (10)$$

The complete formula is as follows:

$$U_i' = F_{scale}(U', S) = S \cdot U' \in \mathbb{R}^{C' \times 1 \times \frac{N'}{2}} \quad (11)$$

3.4.2. Positional encoding

To process ECG signals effectively, we need to encode the positional information and superimpose it onto the embeddings obtained from the previous step. As the waveform of the ECG signal is periodic, the same value will have different meanings in different positions within a period. For example, the peak point of the QRS wave is normal at 0.9 and this value is likely to be an abnormal heartbeat for the peak point of the P wave. We first use a lookup table for positional encoding that can be completed in $O(1)$. This processing time can handle longer sequence lengths during the training. The specific formulas for the sine lookup table are as follows:

$$PE_{(pos, 2i)} = \sin\left(\left(\frac{pos}{10000}\right)^{\frac{2i}{d_{model}}}\right) \quad (12)$$

$$PE_{(pos, 2i+1)} = \cos\left(\left(\frac{pos}{10000}\right)^{\frac{2i}{d_{model}}}\right) \quad (13)$$

where pos is the position and i is the dimension. According to the formulas above, each dimension of the positional encoding corresponds to a sinusoid. The wavelengths form a geometric progression from 2π to $10,000 \cdot 2\pi$. We select this function because it allows the model to learn easily according to the relative positions. For instance, for any fixed offset k , PE_{pos+k} can be represented as a linear function of PE_{pos} .

3.4.3. Lightweight convolution attention (LCA)

In this section, we explain the implementation of the lightweight and contextual attention mechanisms. To make the model lightweight, we apply a novel structure known as LightConv to replace the self-attention in the Transformer model. We first introduce depth-wise convolutions. Each kernel in the depth-wise convolutions operates independently on every channel. The number of parameters can be reduced by d times (d^2 to k), where k is size of Kernel. The output O can be defined as:

$$\begin{aligned} O_{i,c} &= \text{DepthwiseConv}(X, W_c, :, i, c) \\ &= \sum_{j=1}^k W_{c,j} \cdot X \left(i+j-\left\lceil \frac{k+1}{2} \right\rceil, c \right) \end{aligned} \quad (14)$$

where c is output dimension, and weight $W \in \mathbb{R}^{k \times d}$.



Fig. 5. The architecture of Lightweight Convolution attention. We first apply an input projection mapping from dimension d to $2d$, followed by a gated linear unit, and the actual lightweight convolution. SE-NET is also used here as an attention mechanism, but it is worth noting that its role is to provide global attention to the sequence.

LightConv is a depth-wise convolution that shares certain output channels and whose weights are normalized across the temporal dimension using a softmax [49]. Compared to self-attention, LightConv has a fixed context window and determines the importance of context elements with a set of weights that do not change with time. Generally, a common idea in the NLP field is that a structure with common attention is necessary for self-attention models. However, LightConv can also exhibit self-attention parity, and its participation is much less than that of attention. LightConv computes the following formula for the i -th element in the sequence and output channel c :

$$\begin{aligned} \text{LightConv} \left(X, W_{\left\lceil \frac{d}{d} \right\rceil}, :, i, c \right) \\ = \text{DepthwiseConv} \left(X, \text{softmax} \left(W_{\left\lceil \frac{d}{d} \right\rceil}, :, i, c \right) \right) \end{aligned} \quad (15)$$

We divide d channels into H groups and then combine the parameters of each d/H channel, which reduces the number of parameters to d/H . Conventional convolution requires D^2k , and a excision separation convolution has kd , but we only have Hk weight. Softmax is applied to normalize the weights $W \in \mathbb{R}^{k \times H}$.

Figs. 4 and 5 shows the difference between self-attention and LCA architecture. We first place dimensions from d to $2d$, followed by a gated linear unit (GLU) and a lightweight convolution layer.

DropConnect is regularizer for the LightConv. Specifically, we drop the probability p of the normalized weight softmax (W) of each entry divided by $1 - p$ during training. This is equivalent to deleting certain time information in the channel.

In Section 1, we describe the method of strengthening features in the hierarchical structure. Unlike the ordinary RNN structure, in a model with a hierarchical structure, the concept of a channel can actually be understood as the location of the RNN. In our model, we focus on the integrated features of different heartbeat positions by emphasizing the features between the channels of the feature map. In particular, we consider adding a contextual attention mechanism to the LightConv structure to extract the correlation information between heartbeats to improve the recognition rate of the class S .

4. Experiments and evaluation

4.1. Experimental setup

We describe the dataset used in our experiments in Section 3. The dataset includes ten 24-hour bursts of ECG signals collected from a wearable device, including premature ventricular contraction (PVC) and supraventricular premature beat (SPB) signals. First, we do denoising and segmentation and then divide all heartbeats into training set, validation set and test set according to the ratio of 7:1:2 with intra-patient scheme. Second, in order to ensure the validity of the results and avoid random interference, we repeat each experiment ten times, and take the average value as the final result. In particular, we denote three kinds of classes (N, S and V) as three kinds of signals. It can be seen from the Table 1 that there are large gaps in the number of tags of the dataset. Third, to resolve the problem of imbalance in original dataset, we use SMOTE algorithm to resample original dataset so that the possible data

Table 2

The meaning of parameters.

Parameters	Values	Meaning
Batch size	64	Number of heartbeats each time
d_{model}	64	Embedding output size
Num layers	7	Number of attention module
Num heads	7	Number of LightConv in each head
d_{inner}	512	Output dimension of linear layer

imbalance can be removed (the number of all classes equal to the number of class N). Note that only oversampling needs to do on the training data, and SMOTE oversampling is not required to ensure the reliability and generality of the assessment result [51]. We train our model on a machine with one RTX 2080Ti GPU and E5-2690 v3 CPU having 32GB of memory and our code is based on PyTorch.

4.2. Parameter tuning

We first describe the process of model parameter selection as follow. To find the best parameters for model, we use the control variables Method. Four key parameters of the search are d_{model} , layers num, heads num and d_{inner} . We leave three variables unchanged and explore the influence of the last variable on the results. The optimal combination of parameters is as follows: d_{model} is 64, num layers is 7, num heads is 7 and d_{inner} is 512. The meaning of parameters is listed in the Table 2.

4.3. Comparison with rival methods

In order to evaluate the performance of the model, we use the current common metrics: accuracy (Acc), precision (Pre), sensitivity (Sen), and F1 score (F1).

We compare the proposed method with four current mainstream approaches. The data is also obtained from CPSC 2020 and pre-processed using the method of the original paper (segmentation methods and so on). We then compare effect of the proposed model with the model used in the original paper. As some methods do not use the beat segmentation algorithm, instead they use the ready-made annotation, we use the same method [47] for the comparison for fairness. The overall structure has not been changed although some parameters have been adjusted and optimized to make the comparison feasible. The results are shown in the Table 3 and Fig. 6.

As indicated in the table, the current mainstream approaches for heartbeat classification using ECG data that are collected from wearables exhibit varying degrees of weakness. Although these methods performed very well in the MIT-BIH dataset, this is probably owing to the complexity of the data associated with noise and artifact overload. As mentioned previously, problems occur during R-peak detection,

heartbeat segmentation, and classification, especially in the class S detection. Therefore, this is a challenging task because the shape of S is more similar to N than V, whereas most methods are good at classifying V. Our approach improves performance across the board (Fig. 7).

First, the experimental results indicate that the Transformer is more robust and suitable for more complex scenarios. Second, the addition of the attention mechanism has an enhanced effect on countering signal interference.

4.4. Ablation Study

In this section, we evaluate improvements to the original model (Fussing Transformer) from two aspects. First, we evaluate model to see if our LightConv can replace self-attention. Second, we want to know whether the proposed two attention mechanisms work effectively.

4.4.1. Performance improvement

In order to test whether the proposed architecture can improve the detection effect, we conduct the following ablation experiments, as shown in the Table 4 and Fig. 8.

LightConv attention + local attention embedding: The proposed method.

Only LightConv attention: The proposed method without the local attention embedding.

Only LightConv: The proposed method without the local attention embedding and contextual attention.

Fussing Transformer: The Fussing Transformer.

Experimental results have shown that LightConv can be competitive to self-attention mechanism. Thus, self-attention might be an unnecessary structure. Particularly, with the enhancement of LightConv attention mechanism, the ability of the model has been significantly improved, especially for class S. The morphology of Class S is similar to that of class N, but the rhythm features are relatively obvious. As is shown, the application of SENet in this particular convolution structure can also play a similar role in the context compared to global attention. Although there is a phenomenon that the performance of V is not exceeded in the results, we believe it is caused by the small number of abnormal test samples, which is easy to make large errors. At the same time, the Embedding structure with local attention also enhances the rhythmic feature of the heartbeat. We can see some improvement in both types of abnormal beats, especially in the class V, which is also in line with our previous conjecture because the morphology of heartbeat of Class V differs greatly from that of N.

4.4.2. Lightweight

One of our objectives is to provide a lightweight model for ECG wearable devices. For this purpose, we also design a parameter

Table 3

Comparison with rival methods.

Model	Method	N	S	V	Overall (Acc)
The proposed method	Transformer with LightConv attention + CNN embedding	Pre:0.9975 Sen:0.9984 F1:0.9979	Pre:0.9386 Sen:0.8300 F1:0.8810	Pre:0.9158 Sen:0.9447 F1:0.9300	0.9932
Yan [37]	Fussing transformer	Pre:0.9925 Sen:0.9927 F1:0.9925	Pre:0.8739 Sen:0.6931 F1:0.7731	Pre:0.9143 Sen:0.9482 F1:0.9310	0.9728
Jiang [35]	LSTM with attention + CNN embedding	Pre:0.9170 Sen:0.9485 F1:0.9325	Pre:0.8103 Sen:0.7513 F1:0.7797	Pre:0.8698 Sen:0.9081 F1:0.8885	0.9063
Mousavi [52]	BiRNN + CNN embedding	Pre:0.9180 Sen:0.9364 F1:0.9271	Pre:0.8082 Sen:0.7488 F1:0.7774	Pre:0.8577 Sen:0.8853 F1:0.8713	0.8981
Yin [53]	CNNBi-LSTM	– – –	Pre:0.4654 Sen:0.8973 F1:0.6698	Pre:0.5216 Sen:0.9390 F1:0.6111	–

Bold indicates the maximum value in each column of data.

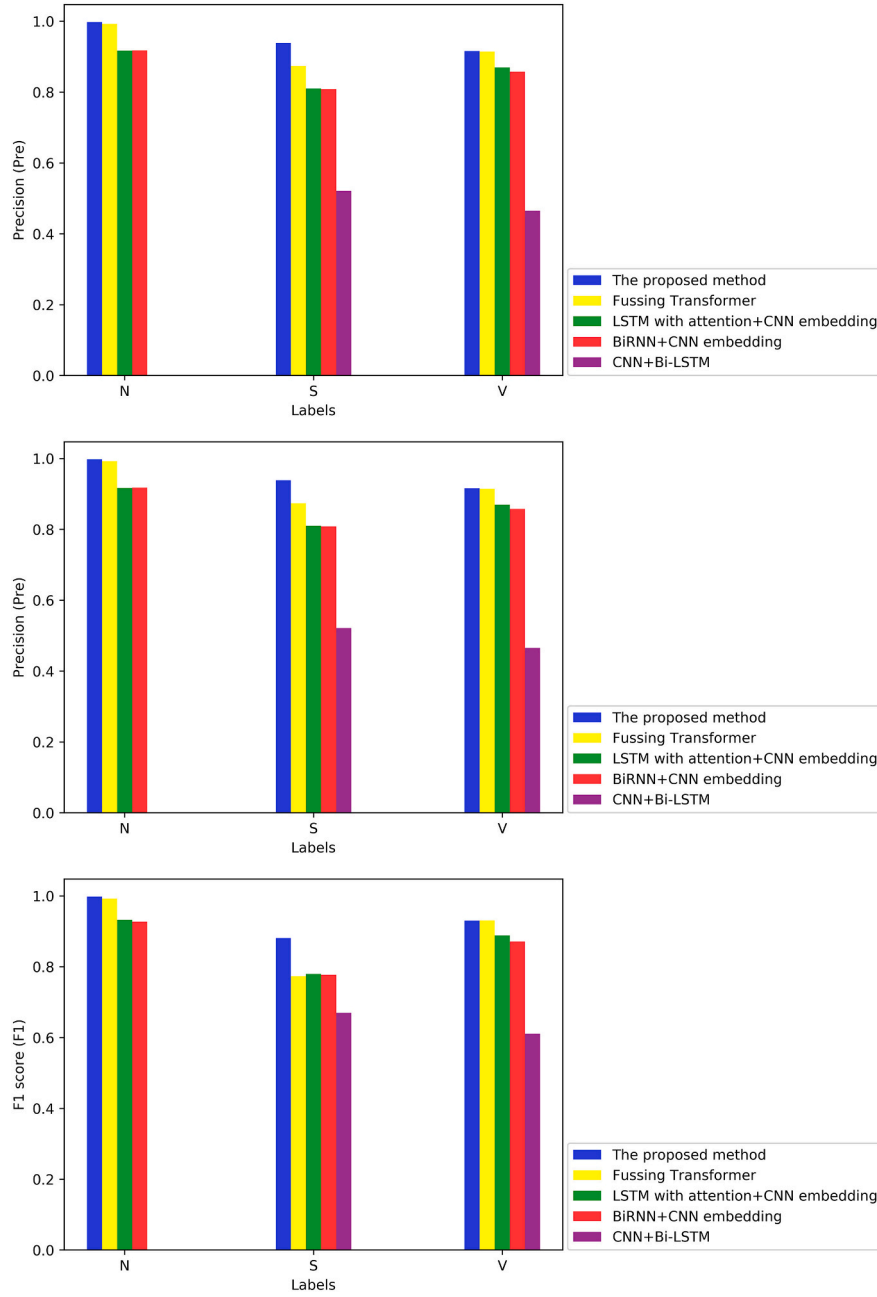


Fig. 6. Comparison of Pre, Sen and F1 between different models based on the same test samples as shown in Table 3.

comparison experiment. Experimental results show that the number of parameters of the LCA structure is 0.0264 M and number of arguments for Fusing Transformer is 0.1114 M. In terms of the overall structure, the number of parameters of the proposed model is 2.69 M while the number of parameters of the original model is 3.26 M. The reason for the small reduction of parameters is that we have improved the input embedding.

Obviously, the parameters of our new LightConv Attention structure is 76.3% less than that of self-attention. On the whole, the parameters of the proposed new structure is 17.5% less than that of the original structure. Therefore, our approaches indicate an exciting result, that is, proposed new structure provides a lightweight reference scheme for future abnormal heartbeat classification models of wearable devices.

4.5. Relationship analysis of principle and performance

We now briefly analyse performance of our new LightConv Attention structure. First, the experimental results demonstrate the power of the Transformer structure in encoding time series data. Our attention structure avoids the shortcomings of the exponential growth of parameters of the self-attention structure and achieves more robust performance because the model uses a hierarchical structure. This is important because timing information plays an important role in the rhythm between beats. In particular, we can observe from the experiments that the attention mechanism that originally plays a role in the hierarchy is still effective in the process of using hierarchy to encode temporal information. To prove this, we visualize the attention mechanism and provide an intuitive example to explain which features are emphasized by the model. The LightConv attention mechanism is illustrated in Fig. 9. According to the different weight values extracted, we directly mark the

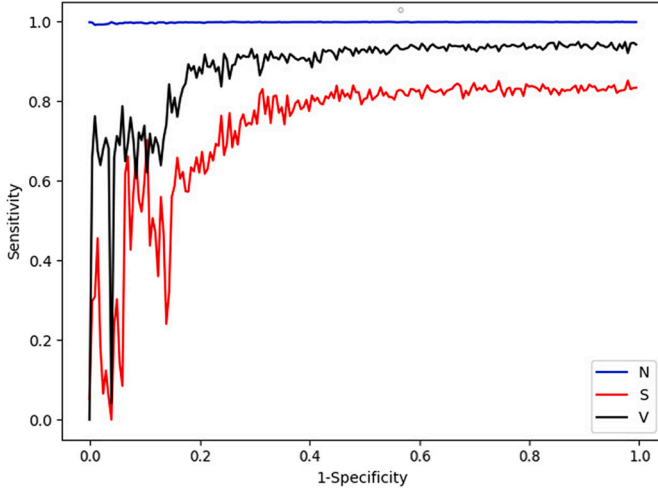


Fig. 7. ROC curve. The area of the lower right represents the AUC intuitively.

Table 4

The experimental results of Ablation Study.

Model	N	S	V	Overall (Acc)
LightConv attention + CNN embedding attention	Pre:0.9975 Sen:0.9984 F1:0.9979	Pre:0.9386 Sen:0.8300 F1:0.8810	Pre:0.9158 Sen:0.9447 F1:0.9300	0.9932
Only LightConv attention	Pre:0.9944 Sen:0.9168 F1:0.9945	Pre:0.9493 Sen:0.7103 F1:0.8126	Pre:0.9027 Sen:0.9082 F1:0.9054	0.9771
Only LightConv	Pre:0.9950 Sen:0.9919 F1:0.9934	Pre:0.9324 Sen:0.7137 F1:0.8086	Pre:0.9112 Sen:0.9681 F1:0.9388	0.9792
Fussing transformer [37]	Pre:0.9925 Sen:0.9927 F1:0.9925	Pre:0.8739 Sen:0.6931 F1:0.7731	Pre:0.9143 Sen:0.9482 F1:0.9310	0.9728

Bold indicates the maximum value in each column of data.

colors of different shades on the heartbeat sequence. Fig. 9 includes 8 N classes heartbeats, with 2 class S heartbeats in the center (5th and 7th). The target is an example of the correct classification of S. Because the LightConv attention mechanism highlights the rhythm feature, both S and N have high weights, although they are very similar in morphology. Interestingly, the heartbeat S is closer to N, so the rhythm information is captured by the attention mechanism.

The experimental results have proved the role of local channel attention in embedding. The embedding of local channel attention is a more effective structure for the Transformer because of the local feature extraction and feature emphasis of the CNN. As shown in Fig. 3, after each convolutional pooling of the model, a channel attention mechanism module is connected, and the weight corresponding to each channel of the pooling feature is calculated. Therefore, each heartbeat through the convolutional network would produce three weights ($weight1 \in \mathbb{R}^{1 \times 32}$, $weight2 \in \mathbb{R}^{1 \times 64}$, $weight3 \in \mathbb{R}^{1 \times 128}$). We select three different classes of heartbeats to visualize the weights. As indicated in Fig. 10, the principle is more obvious during the process of distinguishing V and N, because the local difference of heartbeats between them is large, which is suitable for the use of the local attention mechanism in this study. It is also important to note that it can be clearly observed from the heat map that there are many weights approaching 0 (the heat is approaching white), which demonstrates many detections of arrhythmia in the heartbeat.

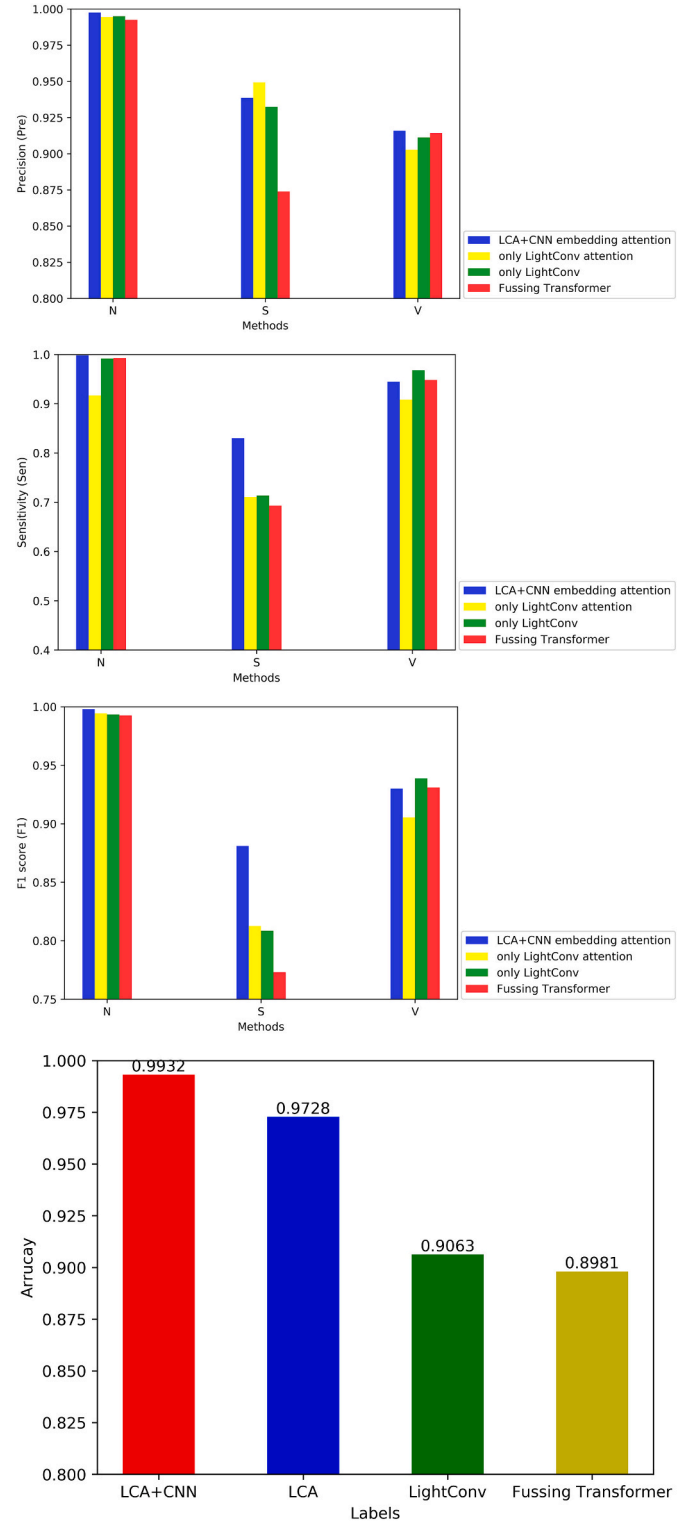


Fig. 8. Comparison of Pre, Sen and F1 between different models in Ablation Study as shown in Table 4.

5. Conclusions and future work

We have proposed the Lightweight Fussing Transformer for arrhythmia detection on a wearable device based on the Fussing Transformer with two attention mechanisms. We use local channel-wise attention to weight intra-heartbeat morphological features learned by CNN embedding. In particular, we replace the original 1D convolutional

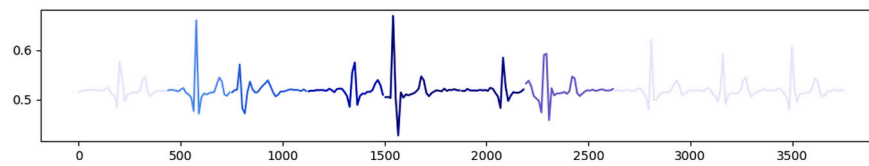


Fig. 9. Visualizing the attention mechanism between heartbeats.

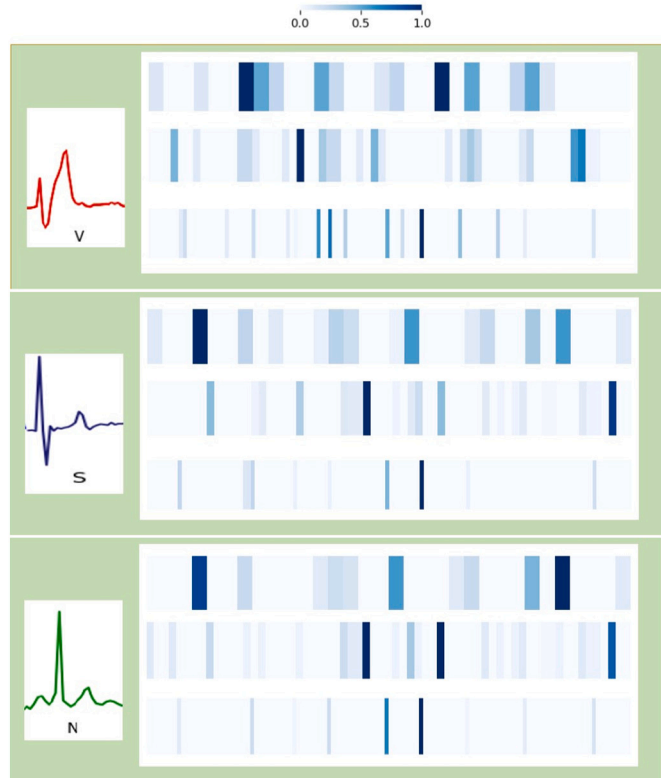


Fig. 10. Visualizing the local attention Mechanism. Each heartbeat corresponds to three weights, from top to bottom: $weight1 \in \mathbb{R}^{1 \times 32}$, $weight2 \in \mathbb{R}^{1 \times 64}$, $weight3 \in \mathbb{R}^{1 \times 128}$.

embedding with an advanced local attention embedding mechanism, which can weigh the feature information that is extracted from the CNN network. To replace self-attention in the Fusing Transformer model, we also implement the novel LightConv attention structure. LCA applies the simplified depth-wise approach and adds SENet for feature weighting between heartbeats. Extensive experiments demonstrate that the performance of the proposed model on the CPSC2020 dataset is outstanding compared to advanced models [35,52] on the MIT-BIH dataset. In the future, we will continue to explore the combination of a lightweight model and wearable dynamic ECG data.

Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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